

Reductions for the finite implication $E677 \models_{\text{fin}} E255$

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May 2026

Abstract

This note records a proof-oriented package of reductions for the finite implication $E677 \models_{\text{fin}} E255$ for magmas. The main output is a route graph: the target law at a point is equivalent to a unique right fixed point, to a left-division identity called Lemma L, and to several finite orbit and collision targets. The orbit package now excludes periods 2 and 3, and isolates a period-four gate. The note also quarantines false trails that should not be reused, including a previously suspected quotient family and a full-shift law.

2020 Mathematics Subject Classification. Primary 08A05; Secondary 08B05, 03C13, 68V15.

Keywords. Finite magma, equational law, finite implication, left division, equational theories.

1 Introduction

A *magma* is a set M equipped with a binary operation $\diamond$. The two laws studied here are

$$\begin{aligned} E677 : \quad & x = y \diamond (x \diamond ((y \diamond x) \diamond y)), \\ E255 : \quad & x = ((x \diamond x) \diamond x) \diamond x. \end{aligned}$$

The goal is the finite implication

$$E677 \models_{\text{fin}} E255,$$

that is, whether every finite magma satisfying $E677$ also satisfies $E255$. The Equational Theories Project (ETP) paper singles out this finite implication, up to duality, together with the dual implication $E2910 \models_{\text{fin}} E47$ [1, Problem 8.1]. The ETP blueprint chapter on Equation 677 contains the local structure theory and the standard finite obstructions used below [4]; the dashboard is a source for cross-checking equation labels [3].

This note is a proof dossier, not a broad search report. The reductions below are organized around a fixed element x of a finite $E677$ -magma. They show that the target law at x is the same as a single right fixed point, that any such fixed-point witness is unique, and that the same target can be expressed as a left-division identity (Lemma L). The later sections turn possible failures of the direct fixed-point route into explicit finite orbit and collision targets.

2 Preliminaries and external input

For $a \in M$, write

$$L_a(t) = a \diamond t, \quad R_a(t) = t \diamond a, \quad S(t) = t \diamond t.$$

If L_a is bijective, write $a \backslash b$ for the unique element u satisfying $a \diamond u = b$. All products in this note are non-associative and are parenthesized when there is any possible ambiguity.

The finite hypothesis first enters through the following standard ETP fact.

Proposition 1 (Left division in finite E677-magmas). *Let M be a finite magma satisfying E677. Then every left translation L_a is bijective. Moreover*

$$a \backslash b = b \diamond ((a \diamond b) \diamond a)$$

for all $a, b \in M$.

Proof. Law E677 gives

$$b = a \diamond (b \diamond ((a \diamond b) \diamond a)).$$

Thus L_a is surjective. Since M is finite, L_a is bijective, and the displayed preimage formula is exactly the formula for $a \backslash b$. $\square$

Lemma 2 (Division calculus). *Let M be a finite magma satisfying E677. Then*

$$a \backslash (a \backslash b) = (a \backslash b) \diamond (b \diamond a), \quad b \backslash (a \backslash b) = (a \diamond b) \diamond a$$

for all $a, b \in M$.

Proof. For the first identity, Proposition 1, applied to $a \backslash b$ in place of b , gives

$$a \backslash (a \backslash b) = (a \backslash b) \diamond ((a \diamond (a \backslash b)) \diamond a) = (a \backslash b) \diamond (b \diamond a),$$

because $a \diamond (a \backslash b) = b$. For the second identity, Proposition 1 gives

$$b \diamond ((a \diamond b) \diamond a) = a \backslash b,$$

so uniqueness of left division by b gives $b \backslash (a \backslash b) = (a \diamond b) \diamond a$. $\square$

The ETP chapter also records several strategic facts that are used here only as proof discipline. The unrestricted implication $\text{E677} \not\models \text{E255}$ is witnessed by the free E677-magma [4, Section 13.2]; therefore any derivation of the finite implication must use finite structure somewhere. Linear models satisfying E677 satisfy E255, and affine-fiber extensions over such models are ruled out by the ETP structure theory [4, Lemmas 13.3–13.4]. The same chapter gives finite E677-magmas that are not right-cancellative [4, Section 13.1]. Hence no argument below may use right cancellation, associativity, an identity element, or group or quasigroup intuition unless such structure has first been derived. The finite-map lemmas in the ETP finite chapter are a natural source of future proof tools [5, Lemmas 5.5–5.7].

Lemma 3 (Transformed identity). *Let M be a finite magma satisfying E677. Then*

$$z = (y \diamond z) \diamond ((y \diamond (y \diamond z)) \diamond y)$$

for all $y, z \in M$.

Proof. Substitute $y \diamond z$ for x in E677. This gives

$$y \diamond z = y \diamond ((y \diamond z) \diamond ((y \diamond (y \diamond z)) \diamond y)).$$

Cancel the common left factor y using Proposition 1. $\square$

Lemma 4 (Key identity). *Let M be a finite magma satisfying E677. Then*

$$(y \diamond (y \diamond z)) \diamond y = z \diamond (((y \diamond z) \diamond z) \diamond (y \diamond z))$$

for all $y, z \in M$.

Proof. Lemma 3 gives

$$z = (y \diamond z) \diamond ((y \diamond (y \diamond z)) \diamond y).$$

On the other hand, applying E677 with the parameter $y \diamond z$ gives

$$z = (y \diamond z) \diamond (z \diamond ((y \diamond z) \diamond z) \diamond (y \diamond z)).$$

Left cancellation by $y \diamond z$ gives the identity. □

3 The orbit package

Fix a finite E677-magma M and an element $x \in M$. Define

$$c_0 = x, \quad c_{k+1} = x \diamond c_k,$$

and let d be the minimal period of this L_x -orbit, so that $c_d = x$ and $c_0, \dots, c_{d-1}$ are pairwise distinct. Indices on the c_k are read modulo d . Put

$$d_k = c_k \diamond x.$$

Lemma 5 (Orbit recurrence). *For every k ,*

$$c_{k-1} = c_k \diamond d_{k+1}.$$

Consequently $d_1 = c_{d-2}$.

Proof. Apply E677 with $y = x$ and the law variable x replaced by c_k :

$$c_k = x \diamond (c_k \diamond ((x \diamond c_k) \diamond x)) = x \diamond (c_k \diamond d_{k+1}).$$

Since also $c_k = x \diamond c_{k-1}$, left cancellation gives the recurrence. Taking $k = 0$ gives $c_{d-1} = x \diamond d_1$, so $d_1 = x \setminus c_{d-1} = c_{d-2}$. □

Lemma 6 (Basic left-division identities). *For every k ,*

$$x \setminus c_k = c_{k-1}, \quad c_k \setminus c_{k-1} = d_{k+1}.$$

If

$$p = x \setminus x = c_{d-1}, \quad q = x \setminus p,$$

then

$$q = d_1 = c_{d-2} = (x \diamond x) \diamond x,$$

and

$$p \diamond (x \diamond x) = d_1, \quad p \setminus d_1 = x \diamond x.$$

Proof. The first identity follows from $c_k = x \diamond c_{k-1}$. The second is precisely the recurrence in Lemma 5. Since $x \diamond c_{d-1} = c_d = x$, one has $p = c_{d-1}$, and then

$$q = x \setminus p = x \setminus c_{d-1} = c_{d-2} = d_1.$$

Also $d_1 = c_1 \diamond x = (x \diamond x) \diamond x$. Finally, Lemma 2, applied with $a = x$ and $b = x$, gives

$$x \setminus p = p \diamond (x \diamond x).$$

The left side is $q = d_1$, and hence $p \diamond (x \diamond x) = d_1$. The last displayed identity is the definition of left division by p . □

Lemma 7 (A fixed point has the forced index). *If $d_j = x$, then $j \equiv d - 2 \pmod{d}$.*

Proof. The condition $d_j = x$ says $c_j \diamond x = x$. Lemma 3 with $y = c_j, z = x$ gives

$$x = x \diamond (x \diamond c_j).$$

On the other hand, E677 with both variables specialized to x gives

$$x = x \diamond (x \diamond d_1).$$

Two left cancellations yield $c_j = d_1 = c_{d-2}$, so the minimality of the orbit period gives the claimed congruence. $\square$

Corollary 8 (No two-cycle in the L_x -orbit). *The period d is never equal to 2. If $d = 1$, then E255 at x holds.*

Proof. If $d = 1$, then $x \diamond x = x$, and so

$$((x \diamond x) \diamond x) \diamond x = (x \diamond x) \diamond x = x \diamond x = x.$$

If $d = 2$, Lemma 5 gives $d_1 = c_{d-2} = c_0 = x$. But Lemma 7 applied with $j = 1$ would give $1 \equiv d - 2 \equiv 0 \pmod{2}$, a contradiction. $\square$

Lemma 9 (No three-cycle in the L_x -orbit). *The period d is never equal to 3. Consequently any point at which the target has not yet been established must have $d \geq 4$.*

Proof. Assume for contradiction that $d = 3$. Put

$$a = c_1 = x \diamond x, \quad p = c_2 = x \setminus x, \quad e = p \diamond x.$$

Since $d_1 = c_{d-2} = c_1$, one has

$$a \diamond x = a. \tag{1}$$

The orbit recurrence gives, for $k = 1$ and $k = 2$,

$$x = a \diamond e, \quad a = p \diamond a. \tag{2}$$

Apply E677 with law variable a and parameter p . Using (2),

$$a = p \diamond (a \diamond ((p \diamond a) \diamond p)) = p \diamond (a \diamond (a \diamond p)).$$

Comparing this with $a = p \diamond a$ and left-cancelling by p gives

$$a \diamond (a \diamond p) = a.$$

Comparing with (1) and left-cancelling by a gives

$$a \diamond p = x. \tag{3}$$

Together with $x = a \diamond e$, this implies $e = p$ by left cancellation by a .

Now apply E677 with law variable x and parameter a . Using (1),

$$x = a \diamond (x \diamond ((a \diamond x) \diamond a)) = a \diamond (x \diamond (a \diamond a)).$$

Comparing with $x = a \diamond e$ and left-cancelling by a gives

$$x \diamond (a \diamond a) = e = p.$$

Since $x \diamond a = p$, left cancellation by x gives

$$a \diamond a = a. \tag{4}$$

Finally, Lemma 3 with $y = a$ and $z = x$, together with (1) and (4), gives

$$x = (a \diamond x) \diamond ((a \diamond (a \diamond x)) \diamond a) = a \diamond ((a \diamond a) \diamond a) = a.$$

This contradicts the minimality of the period, since $c_0 \neq c_1$ when $d = 3$. $\square$

Lemma 10 (Period-four gate). *Assume that the L_x -orbit of x has period 4. Put*

$$a = c_1 = x \diamond x, \quad q = c_2, \quad p = c_3 = x \setminus x, \quad r = q \diamond x.$$

Then

$$x \diamond a = q, \quad a \diamond x = q, \quad a \diamond r = x, \quad x \diamond q = p, \quad p \diamond a = q.$$

Moreover $r \notin \{x, a, q\}$. Thus, with respect to this element r , a period-four proof has two remaining targets: eliminate either $r = p$, or the case where r lies outside the principal L_x -orbit.

Proof. The equality $x \diamond a = q$ is the definition of c_2 , and $a \diamond x = q$ is $d_1 = c_{d-2}$ in the case $d = 4$. The remaining displayed relations are the orbit definition and the orbit recurrence at $k = 1$ and $k = 3$:

$$a \diamond r = x, \quad p \diamond a = q.$$

If $r = x$, then $x = a \diamond r = a \diamond x = q$, contradicting $c_0 \neq c_2$. If $r = a$, then $q \diamond x = a$ and $a \diamond a = x$. Lemma 3, applied with $y = q$ and $z = x$, gives

$$x = (q \diamond x) \diamond ((q \diamond (q \diamond x)) \diamond q) = a \diamond ((q \diamond a) \diamond q).$$

Comparing with $x = a \diamond a$, left cancellation by a gives

$$(q \diamond a) \diamond q = a. \tag{5}$$

Lemma 4, applied with $y = q$ and $z = x$, gives

$$(q \diamond (q \diamond x)) \diamond q = x \diamond (((q \diamond x) \diamond x) \diamond (q \diamond x)).$$

Using $q \diamond x = a$, $a \diamond x = q$, and (5), this becomes

$$a = x \diamond (q \diamond a).$$

Since $a = x \diamond x$, left cancellation by x gives $q \diamond a = x$. Then (5) gives $a = x \diamond q = p$, contradicting $c_1 \neq c_3$.

If $r = q$, then $q \diamond x = q$ and $a \diamond q = x$. Lemma 4, applied with $y = a$ and $z = x$, gives

$$(a \diamond (a \diamond x)) \diamond a = x \diamond (((a \diamond x) \diamond x) \diamond (a \diamond x)).$$

Using $a \diamond x = q$, $q \diamond x = q$, and $a \diamond q = x$, this becomes

$$q = x \diamond (q \diamond q).$$

Since $q = x \diamond a$, left cancellation by x gives $q \diamond q = a$. Applying Lemma 4 with $y = q$ and $z = x$, and using $q \diamond x = q$, now gives

$$x = (q \diamond (q \diamond x)) \diamond q = x \diamond (((q \diamond x) \diamond x) \diamond (q \diamond x)) = x \diamond (q \diamond q) = q,$$

contradicting $c_0 \neq c_2$. $\square$

4 The route graph

The following display is the dependency graph used in the rest of the note. The first row contains equivalent fixed-point targets. The two lower branches describe two proof mechanisms available after the direct fixed-point route: collision lifting and right-collision rigidity.

$$\begin{array}{ccccc}
 \text{E255 at } x & \iff & q \diamond x = x & \iff & \exists u (u \diamond x = x) & \iff & q \backslash x = x \\
 & & \updownarrow & & & & \updownarrow \\
 & & \text{unique witness } q & & & & ((x \diamond x) \diamond ((x \diamond x) \diamond x)) \diamond (x \diamond x) = x \\
 & & & & & & \updownarrow \\
 & & & & & & (q \diamond x) \diamond q = p.
 \end{array}$$

If the fixed point is not established, the two honest remaining branches are

$$\begin{array}{lcl}
 d_i = d_j = e \implies (c_i \diamond e) \diamond c_i = (c_j \diamond e) \diamond c_j & \dashrightarrow & \text{finite counting pressure on } y \mapsto y \diamond e, \\
 x \notin \text{im } R_x \implies y \diamond x = z \diamond x = r & \implies & (y \diamond r) \diamond y = r \backslash x.
 \end{array}$$

The dashed arrow is a missing lemma, not a theorem; the collision branch is now best viewed as a counting problem rather than a direct propagation problem.

5 The fixed-point route

The fixed-point route is sharper than the existential formulation suggests.

Proposition 11 (Unique right fixed-point witness). *Let M be a finite E677-magma and fix $x \in M$. If $u \diamond x = x$, then*

$$u = q = (x \diamond x) \diamond x.$$

Consequently E255 at x is equivalent to the existence of u with $u \diamond x = x$.

Proof. The first half of the proof is the argument already used in Lemma 7. If $u \diamond x = x$, then Lemma 3 gives

$$x = x \diamond (x \diamond u).$$

The specialization $y = x$, $x = x$ in E677 gives

$$x = x \diamond (x \diamond q).$$

Two left cancellations give $u = q$. Since $q = (x \diamond x) \diamond x$, the identity $q \diamond x = x$ is exactly E255 at x . Thus E255 at x is equivalent to the existence of a right fixed point over x , and the witness is forced to be q . $\square$

Proposition 12 (Equivalent local targets). *With the notation above, the following are equivalent:*

1. E255 at x , equivalently $q \diamond x = x$;
2. $q \backslash x = x$;
3. $((x \diamond x) \diamond ((x \diamond x) \diamond x)) \diamond (x \diamond x) = x$;
4. $(q \diamond x) \diamond q = p$.

Proof. The equivalence of (1) and (2) is just uniqueness of left division by q . To identify the left quotient, apply Lemma 3 with $y = x \diamond x$ and $z = x$:

$$x = q \diamond (((x \diamond x) \diamond q) \diamond (x \diamond x)).$$

Therefore

$$q \backslash x = ((x \diamond x) \diamond q) \diamond (x \diamond x),$$

which proves the equivalence of (2) and (3), because $q = (x \diamond x) \diamond x$.

For (4), use Proposition 1:

$$q \backslash x = x \diamond ((q \diamond x) \diamond q).$$

Since $p = x \backslash x$, injectivity of L_x gives

$$x \diamond ((q \diamond x) \diamond q) = x \iff x \diamond ((q \diamond x) \diamond q) = x \diamond p \iff (q \diamond x) \diamond q = p.$$

Together with (1) $\iff$ (2), this proves the claim. $\square$

Definition 13 (Lemma L). For the fixed element x , write $L(x)$ for the identity

$$((x \diamond x) \diamond ((x \diamond x) \diamond x)) \diamond (x \diamond x) = x.$$

By Proposition 12, $L(x)$ is equivalent, in the present finite E677 setting, to E255 at x .

Remark 14. Lemma L is a target, not an established lemma. Because the unrestricted implication $E677 \models E255$ is false [4, Section 13.2], any successful proof of the fixed-point route must use finiteness in a visible way: for example through left bijection, orbit periodicity, a finite-map lemma, or a collision argument. A derivation that simply rewrites E677 into Lemma L without using finite structure should be treated as suspect.

Lemma 15 (Fixed-map and edge-product form). *For the fixed element x , define*

$$H_x(t) = ((x \diamond t) \diamond x), \quad F_x(t) = x \diamond (t \diamond x).$$

Then

$$x \backslash t = t \diamond H_x(t) \tag{E}$$

for every $t \in M$, and

$$F_x(L_x(t)) = L_x(H_x(t)). \tag{Conj}$$

Consequently F_x and H_x have the same finite cycle structure. Moreover E255 at x is equivalent to either of these maps having a fixed point. When fixed points exist, they are unique: the fixed point of F_x is

$$x \backslash q,$$

and the fixed point of H_x is

$$x \backslash (x \backslash q).$$

Thus, if the target fails at x , both maps are fixed-point-free and every cycle of either map has length at least 2.

Proof. Apply E677 with parameter x and law variable t :

$$t = x \diamond (t \diamond ((x \diamond t) \diamond x)) = x \diamond (t \diamond H_x(t)).$$

This is exactly (E). The conjugacy identity is immediate:

$$F_x(L_x(t)) = x \diamond ((x \diamond t) \diamond x) = L_x(H_x(t)).$$

Since L_x is bijective by Proposition 1, (Conj) gives the same cycle structure: L_x carries each H_x -cycle to an F_x -cycle, and L_x^{-1} carries each F_x -cycle back to an H_x -cycle.

If $F_x(z) = z$, then

$$x \diamond (z \diamond x) = z.$$

The left-inverse formula for L_x , applied to z , also gives

$$x \diamond (z \diamond ((x \diamond z) \diamond x)) = z.$$

Left cancellation by x , then by z , yields $(x \diamond z) \diamond x = x$. Hence a right fixed-point witness over x exists, and Proposition 11 gives E255 at x . More precisely, the same proposition gives

$$x \diamond z = q,$$

so any fixed point of F_x must be $z = x \backslash q$. Conversely, if $u \diamond x = x$, set

$$z = u \diamond ((x \diamond u) \diamond x).$$

Then $x \diamond z = u$ by E677 with parameter x , and another use of the same identity gives $z = x \diamond (z \diamond x)$, so $F_x(z) = z$. Taking the forced witness $u = q$, this fixed point is exactly $x \backslash q$. Finally, (Conj) shows that t is fixed by H_x if and only if $L_x(t)$ is fixed by F_x . Therefore the corresponding fixed point of H_x is $x \backslash (x \backslash q)$. $\square$

Remark 16. Identity (E) turns the fixed-point route into a finite dynamics problem. If $H_x(t) = t$, then $x \backslash t = t \diamond t$, equivalently $L_x(S(t)) = t$, which is the last fixed-point form in the ETP Lemma 13.2 list. At a point where the fixed-point target has not yet been established, the forward orbit of any element under H_x must eventually enter a cycle of length greater than 1. A finite proof via this route would have to show that such an H_x -cycle is incompatible with the edge products $x \backslash t = t \diamond H_x(t)$.

Remark 17 (Nontrivial H_x -cycles are not the obstruction). The previous remark should not be read as a global no-cycle target. In the standard linear model on $\mathbb{F}_5$,

$$u \diamond v = 2u - v,$$

one checks directly that E677 and E255 both hold. For fixed x ,

$$H_x(t) = ((x \diamond t) \diamond x) = 3x - 2t.$$

Thus H_x fixes x , while on the other four elements it has a 4-cycle; for example, when $x = 0$, the cycle is

$$1 \mapsto 3 \mapsto 4 \mapsto 2 \mapsto 1.$$

So the finite-map target is not to rule out all nontrivial H_x -cycles. It is to rule out the fixed-point-free H_x -dynamics forced before the target is proved, using the edge products $x \backslash t = t \diamond H_x(t)$ or an actual finite-map identity of the kind used in the ETP finite-map lemmas.

Remark 18 (Finite-map lemmas need a non-tautological input). The finite-map lemmas from the ETP finite chapter apply only after one has produced an identity of the form $f = f \circ f \circ g$ or $f = g \circ f \circ f$ for concrete self-maps of a finite set. The immediate identities

$$L_x = L_x \circ L_x \circ L_x^{-1}, \quad L_x^{-1} = L_x^{-1} \circ L_x^{-1} \circ L_x$$

only recover the tautology $L_x = L_x \circ L_x^{-1} \circ L_x$. They do not involve the right translation R_x , the maps $H_x = R_x \circ L_x$, $F_x = L_x \circ R_x$, or the gate $t \mapsto (t \diamond x) \diamond t$. A future finite-map proof should therefore display a non-tautological identity for one of those maps before invoking Lemma 5.5 or Lemma 5.6.

6 Collision target

The orbit recurrence suggests a collision strategy, but the exact missing step is narrow.

Proposition 19 (Collision lift). *If $d_i = d_j = e$, then*

$$(c_i \diamond e) \diamond c_i = (c_j \diamond e) \diamond c_j. \quad (\text{C})$$

If in addition $i \not\equiv j \pmod{d}$, then

$$c_i \diamond e \neq c_j \diamond e.$$

Proof. Lemma 3 with $y = c_k$ and $z = x$ gives

$$x = d_k \diamond ((c_k \diamond d_k) \diamond c_k).$$

If $d_i = d_j = e$, left cancellation by e yields (C). If also $c_i \diamond e = c_j \diamond e = s$, then (C) becomes $s \diamond c_i = s \diamond c_j$. Left cancellation by s gives $c_i = c_j$, contradicting the minimal period unless $i \equiv j \pmod{d}$. $\square$

Target 20 (Collision-counting target). Turn a nontrivial collision $d_i = d_j = e$ into a finite counting contradiction, or into the fixed point $q \diamond x = x$. More precisely, let

$$F_e = \{y \in M : y \diamond x = e\}, \quad A_e(y) = y \diamond e.$$

Proposition 19, and more generally Proposition 21, show that A_e is injective on F_e . Thus the old direct target $c_i \diamond e = c_j \diamond e$ is not a plausible standalone propagation lemma for a genuine collision: it is precisely what the splitter forbids unless a separate argument has already produced the fixed point. A collision proof should instead find a natural finite set B_e such that

$$A_e(F_e) \subseteq B_e$$

and $|B_e| < |F_e|$, unless $q \diamond x = x$.

The common shortcut “prove $k \mapsto d_k$ injective” is not by itself a proof of the main theorem. It becomes useful only if the argument also connects the resulting injectivity to the forced fixed point $q \diamond x = x$, or if it supplies the counting pressure requested in Target 20.

7 Right-collision constraints

Suppose the fixed-point target fails at x , so $q \diamond x \neq x$. Proposition 11 says that no element u has $u \diamond x = x$. Thus $x \notin \text{im } R_x$. Since M is finite, R_x is not injective, and there exist $y \neq z$ and $r \neq x$ such that

$$y \diamond x = z \diamond x = r.$$

The following proposition records the forced shape of such a right collision.

Proposition 21 (Right-collision rectangles). *Let $y \diamond x = r$. Then*

$$(y \diamond r) \diamond y = r \setminus x. \tag{R}$$

Consequently, if $y \diamond x = z \diamond x = r$ and $y \diamond r = z \diamond r$, then $y = z$.

Proof. Lemma 3 with $y = y$ and $z = x$ gives

$$x = r \diamond ((y \diamond r) \diamond y),$$

which is exactly (R). If $y \diamond r = z \diamond r = s$, then (R) gives $s \diamond y = s \diamond z$. Left cancellation by s gives $y = z$. $\square$

This proposition is used here as a proof constraint. If the fixed-point route has not yet produced $q \diamond x = x$, then right-collision fibers must split under the second probe $t \mapsto t \diamond r$, while all elements in the fiber preserve the same twisted diagonal value $(t \diamond r) \diamond t = r \setminus x$. A proof may use this rigidity for counting or collision lifting.

For bookkeeping, it is often useful to write the left translations as labels. Since all left translations are permutations, one may write

$$a \diamond b = \sigma(a)(b)$$

with $\sigma(a) \in \text{Sym}(M)$. Law E677 is then the labeled- permutation condition

$$\sigma(y)(\sigma(x)(\sigma(\sigma(y)(x))(y))) = x.$$

The labeling condition is not merely a group-theoretic condition on a subgroup of $\text{Sym}(M)$; the labeling map $a \mapsto \sigma(a)$ is part of the data. In fact this labeling map is injective. If $\sigma(y) = \sigma(z)$, choose any x and put $e = y \diamond x = z \diamond x$. The two specializations of E677 give

$$x = y \diamond (x \diamond (e \diamond y)), \quad x = z \diamond (x \diamond (e \diamond z)).$$

Since $L_y = L_z$, the first equality may be rewritten with left factor z . Left cancellation by z , then by x , and finally by e , gives $y = z$. Thus the labeling is injective. In particular, if $y \diamond x = z \diamond x = r$ with $y \neq z$, the two distinct permutations $\sigma(y), \sigma(z)$ agree at x but must separate at r :

$$\sigma(y)(r) \neq \sigma(z)(r), \quad \sigma(\sigma(y)(r))(y) = r \setminus x = \sigma(\sigma(z)(r))(z).$$

The ETP affine-extension obstruction [4, Lemma 13.4] is therefore best used as a warning against importing linear or affine intuition into this proof target.

8 Obstructions and false routes

Several attractive shortcuts are invalid or unsupported.

1. The family

$$d_k \setminus x = c_{k-2} \diamond c_{k-1}$$

should not be used. The old derivation confused $c_{k-1} \diamond x$ with d_k ; by definition it is d_{k-1} . The active repository does not contain a publication-ready finite table witness for this failure, so the present paper records only the derivation error and quarantines the identity.

2. The full-shift law

$$p \diamond d_k = d_{k+1}$$

is also quarantined. It is not a consequence of the trusted local package.

3. A proof of universal idempotence, $d = 1$, or any universal small-period conclusion would contradict known nontrivial positive models from the ETP finite theory. Such a claim should therefore be audited before it is used.
4. A proof that $H_x(t) = ((x \diamond t) \diamond x)$ has no nontrivial cycles is false as a global statement: Remark 17 gives a positive $\mathbb{F}_5$ model with nontrivial H_x -cycles. Only fixed-point-free bad-point dynamics is a live target.
5. A citation of the ETP finite-map lemmas is not by itself a finite proof. Those lemmas require a non-tautological self-map identity; the formal inverse identities for L_x and L_x^{-1} add no new information.
6. A nontrivial collision $d_i = d_j$ does not automatically propagate backwards to $d_{i-1} = d_{j-1}$. Any cascade proof must derive the propagation or avoid it.

9 Proof targets

The work above leaves three precise proof targets.

Target 22 (Fixed-point route). Prove $q \diamond x = x$, equivalently Lemma L from Definition 13, by an argument that uses finite structure explicitly.

Target 23 (Collision route). Resolve Target 20: from $d_i = d_j = e$, derive either the fixed point $q \diamond x = x$ or a finite counting contradiction for the injective splitter $A_e(y) = y \diamond e$ on the fiber $F_e = \{y : y \diamond x = e\}$.

Target 24 (Period-four route). Eliminate the two remaining cases in Lemma 10: $r = p$, and $r \notin \{x, a, q, p\}$. This is the shortest current orbit-level proof target.

A Proof audit checklist

The reductions in this note use left cancellation only after Proposition 1, where finiteness has made each L_a bijective. They do not use right cancellation, associativity, an identity element, or group/quasigroup structure. Future work should state which proof target it attacks and end, if unfinished, with one missing lemma rather than a broad search prescription.

Every new proof attempt should check the following list before being added to the mathematical record.

1. All products are parenthesized.
2. Every cancellation is a left cancellation by a known bijective left translation.
3. Every use of finiteness is named: left bijectivity, orbit periodicity, a finite-map lemma, or pigeonhole reasoning for a self-map.
4. The false quotient family and false full-shift law are not used.
5. Any claimed obstruction or construction contains either a complete finite operation or a structural verification of the relevant identities.

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